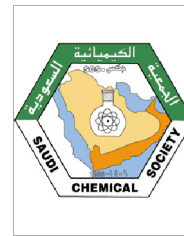




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ORIGINAL ARTICLE

# Biochemical studies on the Natamycin antibiotic produced by *Streptomyces lydicus*: Fermentation, extraction and biological activities



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Received 1 February 2012; accepted 6 April 2012

Available online 26 April 2012

## KEYWORDS

*Streptomyces lydicus*;  
Phylogenetic characteriza-  
tion;  
Fermentation and biological  
activities

**Abstract** Natamycin “polyene” antibiotic was isolated from the fermentation broth of a *Streptomyces* strain No. AZ-55. According to the morphological, cultural, physiological and biochemical characteristics, and 16S rDNA sequence analysis, strain AZ-55 was identified as *Streptomyces lydicus*. It is active in vitro against some microbial pathogens viz: *Staphylococcus aureus*, NCTC 7447; *Bacillus subtilis*, NCTC 1040; *Bacillus pumilus*, NCTC 8214; *Micrococcus luteus*, ATCC 9341; *Escherichia coli*, NCTC 10416; *Klebsiella pneumonia*, NCIMB 9111; *Salmonella typhi* and *Pseudomonas aeruginosa*, ATCC 10145; *S. cerevisiae*, ATCC 9763; *Candida albicans*, IMRU 3669; *Aspergillus flavus*, IMI 111023; *Aspergillus niger*, IMI 31276; *Aspergillus fumigatus*, ATCC 16424; *Fusarium oxysporum*; *Alternaria alternata* and *Rhizoctonia solani*. The active metabolite was extracted using chloroform (1:1, v/v) at pH 7.0. The separation of the active ingredient of the antifungal agent and its purification were performed using both thin layer chromatography (TLC) and column chromatography (CC) techniques. The physico-chemical characteristics of the

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purified antibiotic viz. color, melting point, solubility, elemental analysis (C, H, N, O and S) and spectroscopic characteristics (UV absorbance and IR, mass & NMR spectra) have been investigated. This analysis indicates a suggested empirical formula of  $C_{33}H_{47}NO_{13}$ . The chemical structural analysis with spectroscopic characteristics confirmed that the compound produced by *S. lydicus*, AZ-55 is Natamycin “polyene” antibiotic.

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## 1. Introduction

The search for novel natural products with useful pharmacological activities often includes the isolation of actinomycetes, such as *Streptomyces* species, from soil samples (Ritacco et al., 2003; Sembiring and Goodfellow, 2008). Actinomycetes have been especially useful to the pharmaceutical industry for their seemingly unlimited capacity to produce secondary metabolites with diverse chemical structures and biological activities. Searching for novel actinomycetes constitutes an essential component in the natural product-based drug discovery (Valan Arasu et al., 2008). Actinomycetes are also the focus of attention due to their production of secondary metabolites that may have a range of pharmaceutical and biotechnological applications (Oskay, 2011). *Streptomyces* especially the genus *Streptomyces* were very potent producers of secondary metabolites including antibacterial enzymes and toxins (Ren et al., 2009). Of 10,000 known antibiotics, about 45–55% was produced by *Streptomyces* (Lazzarini et al., 2000). Natamycin, also known as pimaricin, is a polyene macrolide antibiotic, of formula  $C_{33}H_{47}NO_{13}$  produced in submerged cultures of certain *Streptomyces* strains such as *S. natalensis*, *S. chattanoogensis*, and *S. lydicus* (Aparicio et al., 2003). The importance of this antibiotic lies in its broad-spectrum activity against microorganisms, with low toxicity against mammalian cells (Fisher et al., 1978). Natamycin is used to treat bacteria and fungal keratitis (Malecha, 2004). It is especially effective against *Candida*, *Aspergillus*, *Cephalosporium*, *Fusarium* and *Penicillium* (Prajna et al., 2002). Lu et al. (2008) reported the IR spectrum of Natamycin, peaks at  $\nu > 3000\text{ cm}^{-1}$  indicated that there is a typical carboxyl-structure; the  $\nu = 1716\text{ cm}^{-1}$  peak revealed a conjugated ester; the  $\nu = 1570\text{ cm}^{-1}$  peak corresponded to a primary amine; and the  $\nu = 1270\text{--}1150\text{ cm}^{-1}$  peaks showed the existence of different C–O– and the molecular weight at 665. A stereoisomer of 22-(3-Amino-3,6-dideoxy- $\beta$ -D mannopyranosyloxy) 1,3,26-trihydroxy-12-methyl-10-oxo-6, 11,28 trioxatricyclo (22.3.1.0)

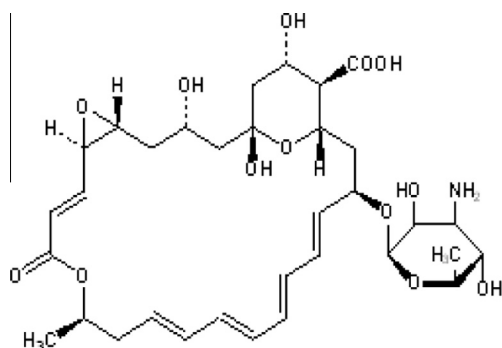


Figure 1 Structure of Natamycin antibiotic.

octacos-8,14,16,18,20-pentaene-25carboxylic acid is shown in (Fig. 1). In the present study, the production of the bioactive substances that demonstrated inhibitory effects against microbial pathogens from *Streptomyces lydicus* was reported, along with some physico-chemical properties of secondary metabolites with high biological activities.

## 2. Materials and methods

### 2.1. Actinomycete strain

Strain AZ-55 was isolated from a suspension of a soil sample (Williams and Davies, 1965) collected from Zagazig district and inoculated onto a starch-nitrate agar. Plates were incubated at 35 °C for 7 days. The isolates were individually maintained on starch-nitrate agar at 4 °C and stored as a mixture of hyphae and spores in 20% glycerol at 80 °C. The selected isolate was allowed to grow in a starch nitrate broth for the purpose of getting a clear supernatant for antimicrobial activity.

### 2.2. Test organisms

#### 2.2.1. Bacteria

2.2.1.1. Gram-positive Bacteria. *Staphylococcus aureus*, NCTC 7447; *Bacillus subtilis*, NCTC 1040; *Bacillus pumilus*, NCTC 8214 and *Micrococcus luteus*, ATCC 9341.

2.2.1.2. Gram-negative Bacteria. *Escherichia coli*, NCTC 10416; *Klebsiella pneumonia*, NCIMB 9111 and *Pseudomonas aeruginosa*, ATCC 10145.

#### 2.2.2. Fungi

2.2.2.1. Unicellular fungi. *C. albicans*, IMRU 3669 and *Saccharomyces cerevisiae* ATCC 9763.

2.2.2.2. Filamentous fungi. *Aspergillus niger*, IMI 31276; *Aspergillus flavus*, IMI 111023; *Aspergillus fumigatus*, ATCC 16424; *Fusarium oxysporum*; *Alternaria alternata*; *Rhizoctonia solani* and *Penicillium chrysogenum*.

### 2.3. Screening for antimicrobial activity

The antimicrobial activity was determined by the cup method assay according to (Kavanagh, 1972).

### 2.4. Taxonomic studies of actinomycete isolate

**Morphological characteristics** of the most potent strain AZ-55 grown on starch nitrate agar medium at 30 °C for 4 days were examined under a scanning electron microscopy (JEOL Technics Ltd.,).

### 2.5. Physiological and biochemical characteristics

The ability of the strain to produce different enzymes was examined by using standard methods. Lecithinase was conducted on egg-yolk medium according to the method of Nitsh and Kutzner (1969); Lipase (Elwan et al., 1977); Protease (Chapman, 1952); Pectinase according to the method of Hankin et al. (1971);  $\alpha$ -amylase according to the method of Cowan (1974) and Catalase test according to the method of Jones (1949); Melanin pigment according to the method of Pridham et al. (1957); Degradation of Esculin and xanthine according to the method of Gordon et al. (1974); Nitrate reduction according to the method of Gordon (1966); Hydrogen sulfide production and oxidase test according to the method of Cowan (1974); utilization of different carbon and nitrogen sources according to the methods of Pridham and Gottlieb (1948); cell wall was performed by the methods of Becker et al. (1964) and Lechevalier and Lechevalier (1970). Cultural characteristics such as color of aerial mycelium, color of substrate mycelium and pigmentation of the selected actinomycete were recorded on ISP agar medium (Shirling and Gottlieb, 1968). Color characteristics were assessed on the scale developed by Kenneth and Deane (1955).

### 2.6. DNA isolation and manipulation

The locally isolated actinomycete strain was grown for 5 days on a starch agar slant at 30 °C. Two ml of a spore suspension was inoculated into the starch-nitrate broth and incubated for 3 days on a shaker incubator at 200 rpm and 30 °C to form a pellet of vegetative cells (pre-sporulation). The preparation of total genomic DNA was conducted in accordance with the methods described by Sambrook et al. (1989).

### 2.7. Amplification and sequencing of the 16S rDNA gene

PCR amplification of the 16S rRNA gene of the local actinomycete strain was conducted using two primers, StrepF; 5.-ACGTGTGCAGCCCAAGACA-3 and StrepR; 5.ACA-AGC CCTGGAAACGGGGT-3., in accordance with the method described by Edwardss et al. (1989). The PCR mixture consisted of 30 pmol of each primer, 100 ng of chromosomal DNA, 200  $\mu$ M dNTPs, and 2.5 units of Taq polymerase, in 50  $\mu$ l of polymerase buffer. Amplification was conducted for 30 cycles of 1 min at 94 °C, 1 min of annealing at 53 °C, and 2 min of extension at 72 °C. The PCR reaction mixture was then analyzed via agarose gel electro-phoresis, and the remaining mixture was purified using QIA quick PCR purification reagents (Qiagen, USA). The 16S rRNA gene was sequenced on both strands via the dideoxy chain termination method, as described by Sanger et al. (1977). The 16S rRNA gene (1.5 kb) sequence of the PCR product was acquired using a Terminator Cycle Sequencing kit (ABI Prism 310 Genetic Analyzer, Applied Biosystems, USA).

### 2.8. Sequence similarities and phylogenetic analysis

The BLAST program ([www.ncbi.nlm.nih.gov/blst](http://www.ncbi.nlm.nih.gov/blst)) was employed in order to assess the degree of DNA similarity. Multiple sequence alignment and molecular phylogeny were evaluated using BioEdit software (Hall, 1999). The phylogenetic tree was displayed using the TREE VIEW program.

### 2.9. Fermentation

A loopful of the *Streptomyces* sp. from the 5-day culture age was inoculated into 250 ml Erlenmeyer flasks containing 75 ml of antibiotic production medium (seven flasks). The flasks were incubated on a rotary shaker (200 rpm) at 30 °C for 5 days. A twenty-liter total volume was filtered through Whatman No. 1 filter paper, followed by centrifugation at 5000 r.p.m for 20 min. The clear filtrates were tested for their activities against the test organisms (Sathi et al., 2001).

### 2.10. Extraction

The clear filtrate was adjusted to different pH values (4–9) and the extraction process was carried out using different solvents separately at the level of 1:1 (v/v). The organic phase was concentrated to dryness under vacuum using a rotary evaporator (Atta, 2010).

### 2.11. Precipitation

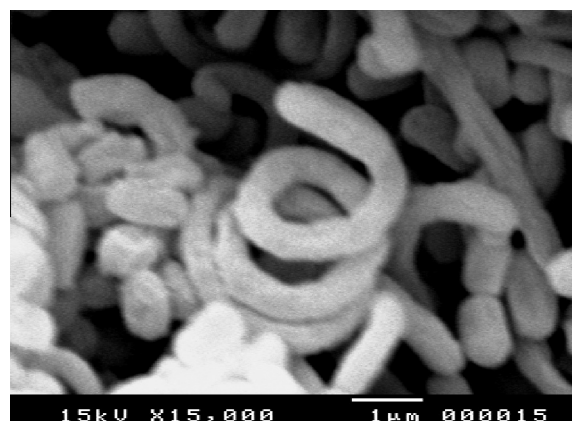
The precipitation process of the crude compound dissolved in the least amount of the solvent was carried out using petroleum ether (b.p 60–80 °C) followed by centrifugation at 5000 r.p.m for 15 min. The precipitate was tested for its antimicrobial activities (Atta et al., 2009).

### 2.12. Separation

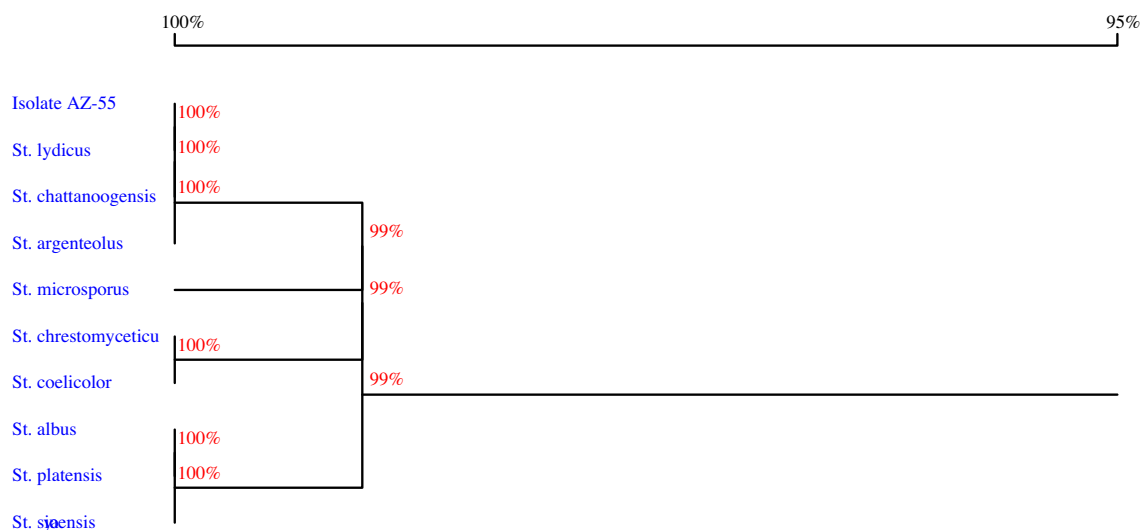
Separation of the antimicrobial agent(s) into its individual components was conducted by a thin layer chromatography using chloroform and methanol (24:1, v/v) as a solvent system (Atta et al., 2009).

### 2.13. Purification

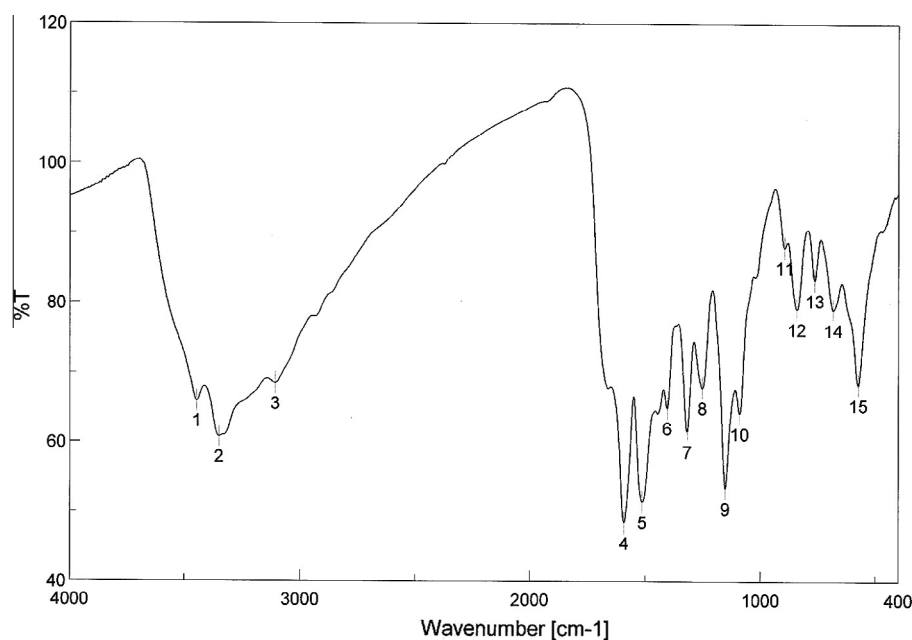
The purification of the antimicrobial agent(s) was carried out using a silica gel column (2  $\times$  25) chromatography. Chloroform–methanol (10:2, v/v/v), was used as an eluting solvent. The column was left overnight until the silica gel (Prolabo) was completely settled. One-ml crude precipitate to



**Plate 1** Scanning electron micrograph of the actinomycete isolate AZ-55 growing on starch nitrate agar medium showing spore chain spiral shape and spore surface smooth (X15,000).



**Figure 2** The phylogenetic position of the local *Streptomyces* sp. strain among neighboring species. The phylogenetic tree was based on the multiple sequence alignment comparisons of 16S rDNA sequences.



**Figure 3** I.R. spectrum of antimicrobial agent produced by *Streptomyces lydicus*, AZ-55.

be fractionated was added on the silica gel column surface and the extract was adsorbed on top of silica gel. Fifty fractions were collected (each of 5 ml) and tested for their antimicrobial activities (Atta et al., 2009).

#### 2.14. Physico-chemical properties of the antimicrobial agent

##### 2.14.1. Elemental analysis

The elemental analysis of C, H, O, N, and S was carried out at the micro analytical center, Cairo University, Egypt.

##### 2.14.2. Spectroscopic analysis

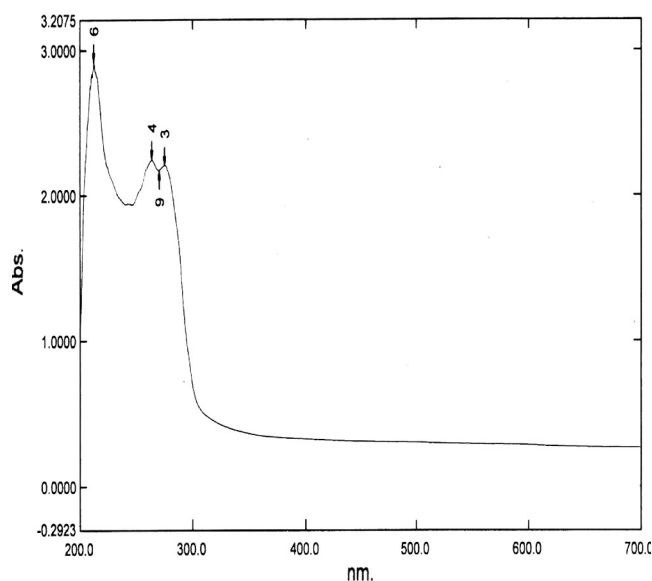
The IR, UV, Mass and NMR spectra were determined at the micro analytical center, Cairo University, Egypt.

##### 2.14.3. Biological activity

The minimum inhibitory concentration (MIC) could be determined by the cup assay method (Kavanagh, 1972).

##### 2.14.4. Characterization of the antimicrobial agent

The antimicrobial agent produced by *Streptomyces* sp. was identified according to the recommended international



**Figure 4** Ultraviolet absorbance of antimicrobial agent produced by *Streptomyces lydicus*, AZ-55.

references of Umezawa (1977), and Berdy (1974, 1980a, 1980b, 1980c), and Eric (1999).

### 3. Results

#### 3.1. Screening for the antimicrobial activities

One of the actinomycete cultures AZ-55 from eight cultures was found to exhibit a wide spectrum of antimicrobial activities (Gram-positive and Gram-negative bacteria and unicellular and filamentous fungi) (Table 1).

#### 3.2. Identification of the actinomycete isolate

##### 3.2.1. Morphological characteristics

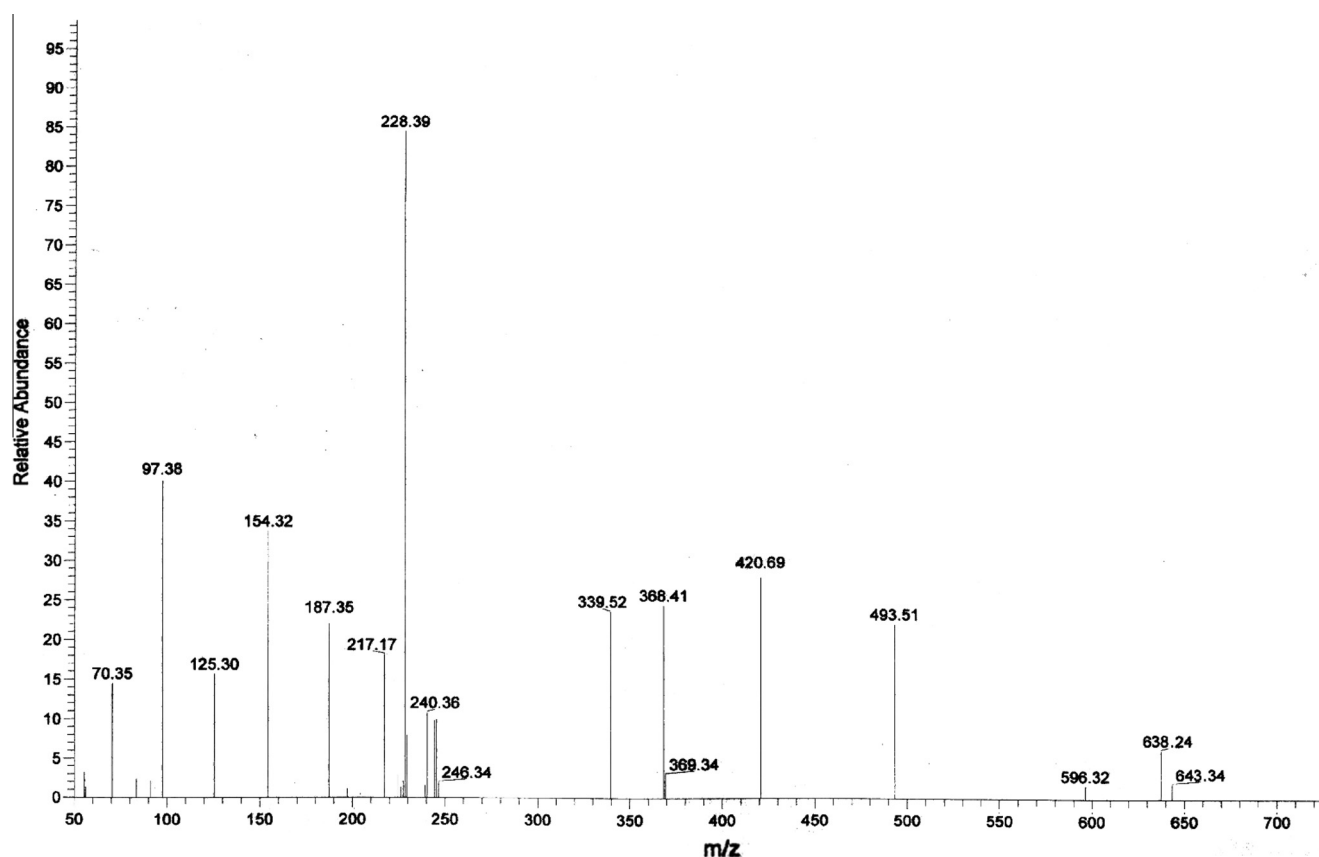
The vegetative mycelia grew abundantly on both synthetic and complex media. The aerial mycelia grew abundantly on starch-nitrate agar medium; oatmeal agar medium (ISP-3) and inorganic salts starch agar medium (ISP-4). The spore chains were spiral, and had a smooth surface (Plate 1). Neither both sclerotic granules and sporangia nor flagellated spores were observed (see Table 2).

##### 3.2.2. Cell wall hydrolysate

The cell wall hydrolysate contains LL-diaminopimelic acid (LL-DAP) and a sugar pattern not detected.

#### 3.3. Color and culture characteristics

As shown in Table 2, the AZ-55 grew on the ISP-media. The isolate exhibited good growth on starch-nitrate agar medium,



**Figure 5** Mass-spectrum of antimicrobial agent produced by *Streptomyces lydicus*, AZ-55.

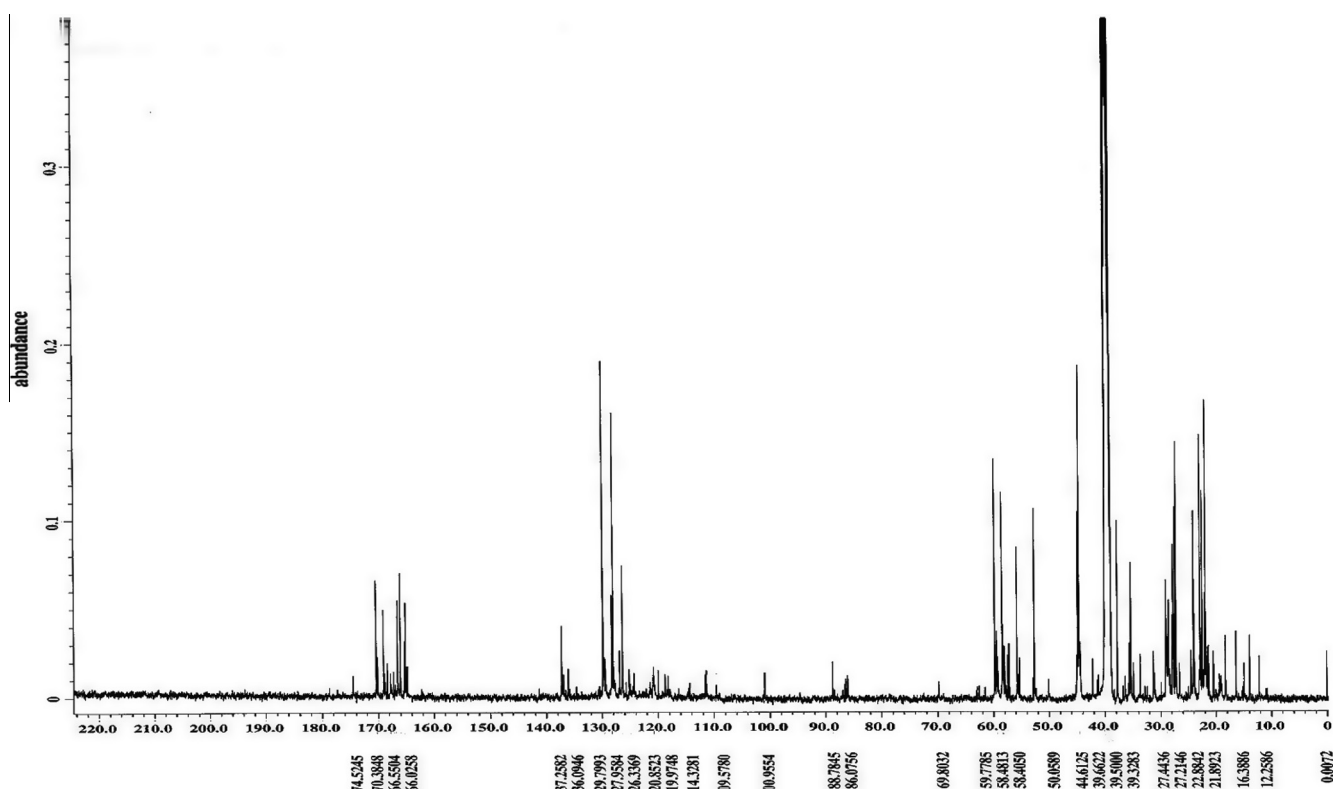


Figure 6 NMR-spectrum of antimicrobial agent produced by *Streptomyces lydicus*, AZ-55.

**Table 1** Antimicrobial activities produced by actinomycete isolate AZ-55.

| Test organisms                     | Mean values of inhibition zones (in mm) |
|------------------------------------|-----------------------------------------|
| <b>A. Bacteria</b>                 |                                         |
| a. Gram positive cocci             |                                         |
| Staph. aureus, NCTC 7447           | 28.0                                    |
| Micrococcus luteus, ATCC 9341      | 29.5                                    |
| b. Gram positive bacilli           |                                         |
| Bacillus subtilis, NCTC 10400      | 28.0                                    |
| Bacillus pumilus, NCTC 8214        | 28.0                                    |
| c. Gram negative bacteria          |                                         |
| Escherichia coli, NCTC 10416       | 27.0                                    |
| Klebsiella pneumonia, NCIMB 9111   | 26.0                                    |
| Pseudomonas aeruginosa, ATCC 10145 | 24.0                                    |
| <b>B. Fungi</b>                    |                                         |
| a. Unicellular fungi               |                                         |
| Candida albicans, IMRU 3669        | 17.0                                    |
| Saccharomyces cerevisiae ATCC 9763 | 17.0                                    |
| b. Filamentous fungi               |                                         |
| Aspergillus niger IMI 31276        | 20.0                                    |
| Aspergillus fumigatus ATCC 16424   | 0.0                                     |
| Aspergillus flavus IMI 111023      | 19.0                                    |
| Alternaria alternata               | 21.0                                    |
| Fusarium oxysporum                 | 21.0                                    |
| Rhizoctonia solani                 | 21.0                                    |
| P. chrysogenum                     | 0.0                                     |

the aerial mycelium showed a dark gray color, substrate mycelium is moderate yellowish brown and the diffusible pigment is dark grayish yellowish brown. No growth on tryptone-yeast extracts broth (ISP-1) and yeast extract-malt extract agar medium (ISP-2). Good growth was detected on oat-meal agar medium (ISP-3). Aerial mycelium is light gray, and substrate mycelium is moderate yellowish brown and no diffusible pigments are seen. Moderate growth was detected on inorganic salts-starch agar medium (ISP-4). Aerial mycelium is light gray and substrate mycelium is light brown and no diffusible pigments are visible. Moderate growth was detected on glycerol-asparagine agar medium (ISP-5). Aerial mycelium is light gray; substrate mycelium is brown and no diffusible pigment. Poor growth was detected on peptone yeast extract-iron agar medium (ISP-6), aerial mycelium is light gray; substrate mycelium is light yellowish brown, and diffusible pigment is moderate brown. Poor growth was detected on tyrosine agar medium (ISP-7), aerial mycelium is light gray, substrate mycelium is light yellowish brown, and diffusible pigment is moderate yellowish brown.

### 3.4. Physiological and biochemical characteristics

The actinomycete isolate AZ-55 could hydrolyze protein, starch, and lecithin and casein hydrolysis is positive whereas lipid, pectin and Catalase tests are negative. Melanin pigment, production of H<sub>2</sub>S, KCN test, and nitrate reduction are negative. Degradation of esculin, xanthine, utilization of citrate, and decomposition of urea are positive. The isolate utilizes, mannose, glucose, galactose, sucrose, mannitol, raffinose, meso-inositol, arabinose, lactose, maltose, fructose, sodium



**Table 2** The morphological, physiological and biochemical characteristics of the actinomycete isolate AZ-55.

| Characteristics                                               | Result                   | Characteristics                                      | Result |
|---------------------------------------------------------------|--------------------------|------------------------------------------------------|--------|
| <i>Morphological characteristics</i>                          |                          |                                                      |        |
| Spore chains                                                  | Spiral                   | Mannitol                                             | ++     |
| Spore mass                                                    | Gray                     | L-Arabinose                                          | +      |
| Spore surface                                                 | Smooth                   | meso-Inositol                                        | ++     |
| Color of substrate mycelium                                   | Yellowish brown          | Lactose                                              | +      |
| Diffusible pigment                                            | Moderate yellowish brown | Maltose                                              | +      |
| Motility                                                      | Non-motile               | D-fructose                                           | +      |
|                                                               |                          | Sodium malonate                                      | +      |
| <i>Cell wall hydrolysate</i>                                  |                          | <i>Utilization of amino acids</i>                    |        |
| Diaminopimelic acid (DAP)                                     | LL-DAP                   | L-Cysteine                                           | +      |
| Sugar pattern                                                 | Not-detected             | L-Valine                                             | —      |
| <i>Physiological and biochemical properties hydrolysis of</i> |                          | L-Histidine                                          | —      |
| Starch                                                        | +                        | L-Phenylalanine                                      | +      |
| Protein                                                       | +                        | L-Arginine                                           | +      |
| Lipid                                                         | —                        | L-Glutamic acid                                      | +      |
| Pectin                                                        | —                        | Growth inhibitors                                    |        |
| Casein and Lecithin                                           | +                        | Sodium azide (0.01)                                  | —      |
| Catalase test                                                 | —                        | Phenol (0.1)                                         | —      |
|                                                               |                          | Thallos acetate (0.001)                              | —      |
| <i>Production of melanin pigment on</i>                       |                          | <i>Growth at different temperatures (°C)</i>         |        |
| Peptone yeast-extract iron agar                               | —                        | 10                                                   | —      |
| Tyrosine agar medium                                          | —                        | 15                                                   | ±      |
| Tryptone yeast extract broth                                  | —                        | 20–45                                                | +      |
| <i>Degradation of</i>                                         |                          | 50                                                   | —      |
| Xanthin                                                       | +                        | <i>Growth at different pH values</i>                 |        |
| Esculin                                                       | +                        | 3–4.5                                                | —      |
| H <sub>2</sub> S Production                                   | —                        | 5–8                                                  | +      |
| Nitrate reduction                                             | —                        | 8.5–12                                               | —      |
| Citrate utilization                                           | +                        | <i>Growth at different concentration of NaCl (%)</i> |        |
| Urea test                                                     | +                        | 1–7                                                  | +      |
| KCN test                                                      | —                        | 10                                                   | —      |
| <i>Utilization of carbon sources</i>                          |                          |                                                      |        |
| D-Xylose                                                      | —                        |                                                      |        |
| D-Mannose                                                     | +                        |                                                      |        |
| D-Glucose                                                     | +                        |                                                      |        |
| D-Galactose                                                   | +                        |                                                      |        |
| Sucrose                                                       | +++                      |                                                      |        |
| L-Rhamnose                                                    | —                        |                                                      |        |
| Raffinose                                                     | +                        |                                                      |        |
| Starch                                                        | +++                      |                                                      |        |

+ = Positive, — = Negative, ± = doubtful results, ++ = moderate growth, and +++ = good growth.

malonate, L-phenylalanine, L-arginine, L-glutamic acid, xylose and L-cysteine whereas it failed to utilize L-valine, histidine and rhamnose. Good growth could be detected within a temperature range of 20–45 °C. Growth in the presence of NaCl up to 7% was recorded and growth at different pH values from 5 to 8 was also recorded and finally no growth in the presence of growth inhibitors; sodium azide (0.01 w/v), phenol (0.1 w/v) and thallos acetate (0.001 w/v) (Table 3).

### 3.5. Taxonomy of actinomycete isolate, AZ-55

This was performed basically according to the recommended international keys viz. (Buchanan and Gibbons, 1974; Williams, 1989; Hensyl, 1994). On the basis of the previously collected data and in view of the comparative study of the recorded properties of AZ-55 in relation to the most closest reference strain, viz. *S. lydicus*, it could be stated that

actinomycetes isolate, AZ-55 is suggestive of being likely belonging to *S. lydicus*, AZ-55 (Table 4).

### 3.6. Amplification of the 16S rDNA gene

The 16S rDNA gene was amplified by polymerase chain reaction (PCR) using the universal primers. The primers used for 16S rDNA sequencing were 16F357 of the sequence strepF; 5'-ACGTGTGCAGCCCAAGACA-3' and strpR; 5'-ACA-AGCCCTGGAAACGGGGT-3', the product of the PCR was analyzed on 1.5% ethidium bromide gel.

### 3.7. Molecular phylogeny of the selected isolate

The 16S rDNA sequence of the local isolate was compared to the sequences of *Streptomyces* spp. In order to determine the relatedness of the local isolate to these *Streptomyces* strains.

**Table 3** Culture characteristics of the actinomycete isolate AZ-55.

| Medium                                           | Growth    | Aerial mycelium                         | Substrate mycelium                                             | Diffusible pigments                                    |
|--------------------------------------------------|-----------|-----------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| 1-Starch nitrate agar medium                     | Good      | 266-d.gray<br>Dark gray                 | 77-m.ybr<br>Moderate yellowish brown                           | 81-d.gy-ybr<br>Dark grayish yellowish brown            |
| 2-Tryptone yeast extract broth (ISP-1)           | No growth | –                                       | –                                                              | –                                                      |
| 3-Yeast extract malt extract agar medium (ISP-2) | No growth | –                                       | –                                                              | –                                                      |
| 4-Oatmeal agar medium (ISP-3)                    | Good      | 264-l. gray<br>Light gray               | 77-m.ybr<br>Moderate yellowish brown                           | –                                                      |
| 5-Inorganic salts starch agar medium (ISP-4)     | Moderate  | 264-l. gray                             | 57-l.br                                                        | –                                                      |
| 6-Glycerol-asparagine agar medium (ISP-5)        | Good      | Light gray<br>264-l. gray               | Light brown<br>57-l.br                                         | –                                                      |
| 7-Peptone yeast extract iron agar medium (ISP-6) | Poor      | Light gray<br>264-l. gray               | Light brown<br>77-m.ybr                                        | 58 m-brown                                             |
| 8-Tyrosine agar medium (ISP-7)                   | Poor      | Light gray<br>264-l. gray<br>Light gray | Moderate yellowish brown<br>76-l-y-br<br>Light yellowish Brown | Moderate brown<br>77-m.ybr<br>Moderate yellowish brown |

The color of the organism under investigation was consulted with the ISCC-NBS color-name charts illustrated with centroid color.

The phylogenetic tree (as displayed by the Tree View program) revealed that the locally isolated strain is closely related to *Streptomyces* sp., rather related to *Streptomyces* sp., rather than to *S. lydicus* (Fig. 1). Multiple sequence alignment was conducted on the sequences of the 16S rDNA gene of *S. lydicus*. Computer assisted DNA searches against bacterial database similarly revealed that the 16S rDNA sequence was 99% identical to *S. lydicus* (Fig. 1).

### 3.8. Fermentation and purification of the antimicrobial agent

The fermentation process was carried out for 3 days at 30 °C using the liquid starch nitrate medium as production medium. Filtration was conducted followed by centrifugation at 5000 r.p.m. for 15 min. The clear filtrates containing the active metabolite (20 l), were adjusted to pH 7.0 then the extraction process was carried out using chloroform at the level of 1:1 (v/v). The organic phase was collected, and evaporated under reduced pressure using a rotary evaporator. The residual material was dissolved in the least amount of DMSO and filtered. The filtrates were tested for their antimicrobial activities. The antimicrobial agent was precipitated by petroleum ether (b.p. 60–80 °C) and centrifuged at 5000 r.p.m for 15 min. Separation of the antimicrobial agent(s) into individual components was carried out by a thin-layer chromatography using a solvent system composed of chloroform and methanol (24:1, v/v). Among three bands developed, only one band at  $R_f$  0.7 showed antimicrobial activity. The purification process through column chromatography packed with silica gel indicated that the most active fractions against the tested organisms ranged from 16–25.

### 3.9. Physicochemical characteristics of the antimicrobial agent

The purified antimicrobial agent produces a characteristic odor, its melting point is 184–185 °C. The compound is freely

soluble in chloroform, ethyl acetate, n-butanol, acetone, ethyl alcohol, methanol and 10% isopropyl alcohol, but insoluble in water, petroleum ether, hexane and benzene.

### 3.10. Elemental analysis of the antimicrobial agent

The elemental analytical data of the antimicrobial agent(s) revealed the following: C 59.54%, H 7.12%, N 2.10%, O 31.24% and  $S = 0.0$ . This analysis indicates a suggested empirical formula of  $C_{33}H_{47}NO_{13}$ .

### 3.11. Spectroscopic characteristics of the antimicrobial agent

The infrared (IR) spectrum indicated that there is a typical carboxyl-structure; the  $\nu = 1716\text{ cm}^{-1}$  peak revealed a conjugated ester; the  $\nu = 1570\text{ cm}^{-1}$  peak corresponded to a primary amine; and the  $\nu = 1270\text{--}1150\text{ cm}^{-1}$  peaks showed the existence of different C–O– (Fig. 2). The ultraviolet (UV) absorption spectrum is recorded at a maximum absorption peak at 212.5, 271.0 and 280 nm (Fig. 3). The Mass spectrum showed that the molecular weight was 664.34 (Fig. 4). The NMR spectrum showed that the peaks of  $\delta = 180$  and 165 are indicative of the existence of carboxyl and ester groups, the peak of  $\delta = 130$  was produced by polyethylene (Fig. 5).

### 3.12. Biological activities of the antimicrobial agent

Data of the antimicrobial agent spectrum indicated that the agent is active against Gram positive and Gram negative bacteria and unicellular and filamentous fungi (Table 5). The MIC of the antimicrobial antibiotic was determined and the results showed that the minimum inhibitory concentration (MIC) of the compound against *S. aureus*, NCTC 7447, *Micrococcus lutea*, ATCC 9341 and *B. subtilis*, NCTC 10400 was 1.73 µg/ml,



**Table 4** A comparative study of the characteristics of actinomycete isolate, AZ-55 in relation to reference strain, *Streptomyces lydicus* (C.F. Hensyl, 1994, Page 693 and Table 27.5).

| Characteristics                      | AZ-55           | Hensyl (1994) <i>Streptomyces lydicus</i> |
|--------------------------------------|-----------------|-------------------------------------------|
| <i>Morphological characteristics</i> |                 |                                           |
| Spore mass                           | Gray            | Gray                                      |
| Spore surface                        | Spiral          | Spiral                                    |
| Color of substrate mycelium          | yellowish-brown | yellowish-brown                           |
| Spore surface                        | Smooth          | Smooth                                    |
| Motility                             | Non-motile      | Not-motile                                |
| <i>Cell wall hydrolysate</i>         |                 |                                           |
| Diaminopimelic acid (DAP)            | LL-DAP          | LL-DAP                                    |
| Sugar pattern                        | Not-detected    | Not-detected                              |
| Melanin pigment                      |                 |                                           |
| Hydrolysis of:                       |                 |                                           |
| Casein                               | +               | +                                         |
| Protein                              | +               | +                                         |
| Pectin                               | —               | —                                         |
| Starch                               | +               | +                                         |
| Egg-yolk                             | +               | +                                         |
| <i>Degradation of:</i>               |                 |                                           |
| Esculine                             | +               | +                                         |
| Xanthine                             | +               | +                                         |
| H <sub>2</sub> S production          | —               | —                                         |
| Nitrate reduction                    | —               | —                                         |
| <i>Utilization of:</i>               |                 |                                           |
| Sucrose                              | +               | +                                         |
| Mannitol                             | +               | +                                         |
| meso-Inositol                        | +               | +                                         |
| Rhamnose                             | —               | —                                         |
| L-Cysteine                           | +               | +                                         |
| L-valine                             | —               | —                                         |
| L-phenylalanine                      | +               | +                                         |
| L-Histidine                          | —               | —                                         |
| Optimum growth temperature           | 30 °C           | 25 °C                                     |
| Optimum pH                           | 7               | 7                                         |
| Growth at NaCl (7.0%)                | +               | +                                         |
| <i>Growth inhibitors</i>             |                 |                                           |
| Sodium azide (0.01)                  | —               | —                                         |
| Phenol (0.1)                         | —               | —                                         |
| Thallous acetate (0.001)             | —               | —                                         |

+ = Positive, — = Negative.

for *K. pneumonia*, NCIMB 9111, was 3.9 µg/ml, for *E. coli*, NCTC 10416 and *S. typhi* was 7.8 µg/ml, for *Asp. flavus*, IMI 111023; *F. oxysporum*; *R. solani* and *A. alternata* was 31.25 µg/ml and for *C. albicans*, IMRU 3669 and *S. cerevisiae* was 62.5 µg/ml (see Fig. 6).

### 3.13. Identification of the antimicrobial agent

On the basis of the recommended keys for the identification of antibiotics and in view of the comparative study of the recorded properties of the antimicrobial agent, it could be stated that the antimicrobial agent is suggestive of being belonging to Natamycin “polyene” antibiotic (Table 6).

## 4. Discussion

The increase in the frequency of multi-resistant pathogenic bacteria has created an urgent demand in the pharmaceutical

industry for more rational approaches and strategies to the screening of new antibiotics with a broad spectrum of activity, which resist the inactivation processes exploited by microbial enzymes (Motta et al., 2004). Only one actinomycete culture AZ-55 from eight cultures was found to exhibit a wide spectrum of antimicrobial activities. Identification process has been carried out according to Williams (1989) and Hensyl (1994). For the purpose of identification of actinomycete isolate, the morphological characteristics and microscopic examination emphasized that the spore chain is spiral. Spore mass is light gray; while spore surface is smooth, substrate mycelium is yellowish brown and no diffusible pigment was produced on ISP-media. The results of physiological and biochemical characteristics and cell wall hydrolysate of actinomycetes isolate, exhibited that the cell wall containing LL-diaminopimelic acid (DAP) and sugar pattern of cell wall hydrolysate could not be detected. These results emphasized that the actinomycetes isolate was related to a group of *Streptomyces*. In view of all

**Table 5** Biological activities (MIC) of the antimicrobial agent by paper method assay.

| Test organisms                     | MIC (µg/ml) concentration                                   |
|------------------------------------|-------------------------------------------------------------|
|                                    | Antimicrobial agent produced by Streptomyces lydicus, AZ-55 |
| <b>A. Bacteria</b>                 |                                                             |
| a. <i>Gram positive cocci</i>      |                                                             |
| Staph. aureus, NCTC 7447           | 1.73                                                        |
| Micrococcus luteus, ATCC 9341      | 1.73                                                        |
| b. <i>Gram positive bacilli</i>    |                                                             |
| Bacillus subtilis, NCTC 10400      | 1.73                                                        |
| c. <i>Gram negative bacteria</i>   |                                                             |
| Escherichia coli, NCTC 10416       | 3.9                                                         |
| Klebsiella pneumonia, NCIMB 9111   | 7.8                                                         |
| Salmonella typhi                   | 7.8                                                         |
| <b>B. Fungi</b>                    |                                                             |
| a. <i>Unicellular fungi</i>        |                                                             |
| Saccharomyces cerevisiae ATCC 9763 | 62.5                                                        |
| Candida albicans, IMRU 3669        | 62.5                                                        |
| b. <i>Filamentous fungi</i>        |                                                             |
| Asp. niger, IMI 31276              | 31.25                                                       |
| Alternaria alternate               | 31.25                                                       |
| Fusarium oxysporum                 | 31.25                                                       |
| Rhizoctonia solani                 | 31.25                                                       |

**Table 6** A comparative study of the characteristic properties of the antimicrobial agent in relation to reference Natamycin (polyene) antibiotic.

| Characteristic            | Purified antibiotic                        | Natamycin antibiotic                       |
|---------------------------|--------------------------------------------|--------------------------------------------|
| 1. Melting point          | 184–185 °C                                 | ND                                         |
| 2. Molecular weight       | 664.34                                     | 665.72                                     |
| 3. Chemical analysis (%): |                                            |                                            |
| C                         | 59.54                                      | 59.54                                      |
| H                         | 7.12                                       | 7.12                                       |
| N                         | 2.10                                       | 2.10                                       |
| O                         | 31.24                                      | 31.24                                      |
| S                         | 0.0                                        | 0.0                                        |
| 4. Ultra violet           | 212.5, 271.0 and 280 nm                    | 271, 281, 291 and 305                      |
| 5. Formula                | $\text{C}_{33}\text{H}_{47}\text{NO}_{13}$ | $\text{C}_{33}\text{H}_{47}\text{NO}_{13}$ |

ND = No data.

the previously recorded data, the identification of actinomycetes isolate AZ-55 was suggestive of being belonging to *S. lydicus*, AZ-55. The resulted sequence was aligned with available almost complete sequence of strain types of family streptomycetaceae. It formed a phyloetic line that was closely related to *S. lydicus*, AZ-55, sharing of 16S rDNA gene similarity matrix is 99%.

The active metabolites were extracted by chloroform at pH 7.0. Similar results were obtained by Atta et al. (2011). The purification process through a column chromatography packed with silica gel and an eluting solvent composed of chloroform-methanol (10:2, v/v), indicated that fraction activities were recorded from fraction Nos. 16–25. Many workers used a column chromatography packed with silica gel. Similar results were obtained by El-Naggar (2007) and Atta et al. (2009). The physico-chemical characteristics of the purified antibiotic revealed that, its melting point is 184–185 °C. The compound is freely soluble in chloroform, ethyl acetate,

n-butanol, acetone, ethyl alcohol, methanol and 10% isopropyl alcohol, but insoluble in water, petroleum ether, hexane and benzene. Similar results were recorded by Wenli et al. (2008). A study of the elemental analysis of the antifungal agent with C 59.44%, H 7.22%, N 2.10%, O 31.24% and S = 0.0 lead to an empirical formula of  $\text{C}_{33}\text{H}_{47}\text{NO}_{13}$ . The spectroscopic characteristics of the antimicrobial agent under study revealed the IR spectrum, the peaks at  $\nu > 3000\text{ cm}^{-1}$  indicated that there is a typical carboxyl-structure; the  $\nu = 1716\text{ cm}^{-1}$  peak revealed a conjugated ester; the  $\nu = 1570\text{ cm}^{-1}$  peak corresponded to a primary amine; and the  $\nu = 1294\text{--}1116\text{ cm}^{-1}$  peaks showed the existence of different C–O–. The UV spectrum showed that the active compound dissolved in methanol–water presented four typical absorbance peaks at wavelengths of 212.5, 271.0 and 280 nm, which is the typical characteristic of conjugated polyene chemicals. The mass spectroscopy revealed that the molecular weight is 664.34. Similar results were recorded by Lu et al. (2008). The

MIC of the antimicrobial antibiotic was determined and the results showed that the minimum inhibitory concentration (MIC) of the compound against *S. aureus*, NCTC 7447, *Micrococcus lutea*, ATCC 9341 and *B. subtilis*, NCTC 10400 was 1.73 µg/ml, for *K. pneumonia*, NCIMB 9111, was 3.9 µg/ml, for *E. coli*, NCTC 10416 and *S. typhi* was 7.8 µg/ml, for *Asp. flavus*, IMI 111023; *F. oxysporum*; *R. solani* and *A. alternata* was 31.25 µg/ml and for *C. albicans*, IMRU 3669 and *S. cerevisiae* was 62.5 µg/ml. Similar investigations and results were attained by Kavitha and Vijayalakshmi (2007) and Atta (2010).

Identification of the antimicrobial agent according to the recommended international keys indicated that the antibiotic is suggestive of being Natamycin “polyene” antibiotic (Eric, 1999; Lu et al., 2008).

## 5. Conclusion

Actinomycetes are producers of potent metabolic compounds used commercially as antibiotics and other novel drugs. The present study shows the present data focusing on obtaining microbial local isolates which have the ability to produce an antimicrobial agent. An interesting scope for further research would be to improve the antimicrobial agent production by *S. lydicus*, AZ-55 against pathogenic microorganisms (Gram positive and Gram negative bacteria and unicellular and filamentous fungi). The Natamycin “polyene” antibiotic produced by *S. lydicus*, AZ-55 demonstrated obvious inhibitory effects against microbial pathogens.

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